A

Project Thesis on

STUDENT SKILL EXCHANGE PLATFORM USING MACHINE LEARNING

Submitted in partial fulfillment of the requirements for the award of

DIPLOMA IN COMPUTER ENGINEERING

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With gratitude,

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The thesis entitled **"STUDENT SKILL EXCHANGE PLATFORM USING MACHINE LEARNING"** is a record of bonafide work carried out by us, in partial fulfillment for the award of the Diploma from State Board of Technical Education and Training, Mangalagiri, Andhra Pradesh. The results embodied in this thesis have not been submitted to any other university or institute for the award of any diploma.

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ABSTRACT

In modern educational environments, many students face barriers such as hesitation, fear of judgment, or anxiety in asking questions during classroom sessions. These barriers often prevent them from interacting with faculty members or peers, limiting their learning opportunities and skill development. The **Student Skill Exchange Platform** is a web-based application designed to address these challenges by enabling **peer-to-peer collaboration** in a structured, intelligent, and user-friendly environment.

The platform allows students to **register and create profiles**, including personal details, academic information, and a list of skills they possess or wish to learn. Using **Machine Learning algorithms**, specifically **TF-IDF (Term Frequency – Inverse Document Frequency)** and **Cosine Similarity**, the system automatically analyzes student skill sets to provide **intelligent peer recommendations**. This ensures that each student is matched with peers having similar or complementary skills, promoting **effective and safe knowledge exchange** without the fear of judgment.

Key features of the platform include:

Registration and Login: Secure access for students to create and manage their profiles.

Skill Matching and Peer Recommendation: ML-powered algorithm for recommending suitable peers based on TF-IDF and Cosine Similarity analysis of skill sets.

Request System: Students can **send requests to peers**, which can be accepted or ignored based on preference.

Notifications and Triggers: Automated alerts inform students about received requests, accepted requests, or ignored requests, ensuring smooth communication flow.

Chat Sessions: Once matched or accepted, students can communicate in real-time through the integrated chat module, facilitating discussion, doubt clearing, and collaborative learning.

The platform aims to **empower students with hesitation or fear barriers** by creating a safe and supportive learning environment where knowledge can be shared freely. It also encourages **peer-to-peer engagement**, fosters collaboration, and enhances practical skill development. By leveraging **Machine Learning for intelligent matching**.

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CHAPTER 1

INTRODUCTION

1.1 Background work of the project

In the modern educational ecosystem, skill development plays a crucial role in shaping a student's academic and professional journey. While colleges provide structured academic knowledge through lectures and curriculum-based teaching, many essential practical skills such as programming, communication, graphic design, electronics, video editing, content creation, data analysis, and public speaking are often learned independently by students. These skills are highly valuable in today's competitive world, yet there is limited structured support within institutions to help students effectively share, exchange, and enhance these abilities.

Key Observations in the Current Educational System:

- Many students develop practical skills outside the formal classroom environment.
- There is no organized system inside colleges for structured peer skill sharing.
- Skill exchange mainly depends on informal networks and personal contacts.

. Over time, this hesitation can reduce self-confidence and limit skill growth. Another significant issue observed in many educational institutions is the psychological barrier faced by students. A considerable number of students experience hesitation, fear of judgment, lack of confidence, or anxiety while approaching faculty members to clarify doubts. They may worry about asking "basic" questions, fear criticism, or feel uncomfortable speaking in formal classroom environments. As a result, their doubts remain unresolved, which negatively impacts their learning process and academic performance

- **Psychological Barriers Faced by Students:**
  - Hesitation in asking doubts in front of others.
  - Fear of negative judgment from faculty or classmates.
  - Lack of confidence in expressing questions.
  - Anxiety in formal academic environments.

At the same time, many students are comfortable interacting with their peers. Peer learning creates a friendly, informal, and supportive environment where students can freely express doubts without fear. However, despite this potential, there is no organized system within most colleges to help students identify peers who possess the specific skills they want to learn. Students often depend on personal networks or word-of-mouth, which is inefficient and unreliable.

To address these challenges, the "**Student Skill Exchange Platform"** is designed as a structured, intelligent, and technology-driven solution that enables students to connect based on their skills and learning interests. The primary goal of this platform is to eliminate hesitation barriers and create a safe academic environment where students can learn comfortably from fellow students. By encouraging peer-to-peer interaction, the system reduces fear, builds confidence, and promotes collaborative growth.

- **Core Objectives of the Platform:**
  - To create a structured peer-to-peer learning ecosystem.
  - To reduce hesitation and fear of judgment among students.
  - To enable skill-based intelligent matching between learners and mentors.
  - To promote collaborative academic and professional growth.

The platform integrates Machine Learning algorithms with modern web technologies to create an automated recommendation system. Instead of manually searching for suitable peers, students receive intelligent suggestions based on similarity analysis of their skill sets, interests, and learning requirements. This intelligent pairing mechanism ensures that students are matched efficiently with peers who can teach or learn from them, thereby maximizing knowledge exchange.

Furthermore, this project demonstrates how advanced technologies such as Artificial Intelligence and Machine Learning can be effectively implemented within educational institutions to enhance student engagement. It shifts the focus from traditional one-directional teaching models to collaborative, student-centered learning environments.

In conclusion, the Student Skill Exchange Platform not only promotes skill development but also addresses emotional and psychological barriers like hesitation and fear of judgment. By

fostering a supportive peer-learning ecosystem, the system empowers students to learn confidently, share knowledge freely, and grow both academically and professionally.

1.2 Project Overview

The **Student Skill Exchange Platform** is a web-based intelligent application developed to facilitate peer-to-peer learning among students within an educational institution. The primary aim of the system is to connect students based on their skill sets and learning interests, enabling structured knowledge sharing and collaborative growth.

In many colleges, students possess diverse technical and non-technical skills such as programming, web development, communication, graphic design, electronics, content creation, and data analysis. However, there is no centralized platform that allows students to easily discover peers who can help them learn new skills or collaborate on projects. As a result, students often rely on informal methods like friend circles or social media groups to find collaborators, which can be inefficient and limited in reach.

The Student Skill Exchange Platform addresses this challenge by integrating Machine Learning techniques into a web application. Students can register on the platform by entering their personal details, branch, academic year, and skill set. The system then analyzes the entered skills using **TF-IDF (Term Frequency – Inverse Document Frequency)** to convert textual skill descriptions into numerical representations. After vectorization, **Cosine Similarity** is applied to calculate similarity scores between students' skill profiles. Based on these scores, the system recommends the most suitable students for skill exchange.

The platform also includes filtering options based on branch and academic year, allowing users to refine their search results. Additionally, an integrated chat feature enables direct communication between matched students, making collaboration more convenient and effective.

Technically, the project combines:

- Front-end web technologies for user interaction.
- Backend processing for handling data and implementing algorithms.
- Database management for storing user information.

- Machine Learning algorithms for intelligent recommendation.

This project demonstrates the practical application of Artificial Intelligence concepts in the educational domain. It transforms a simple student directory into an intelligent recommendation system that promotes collaboration, reduces skill gaps, and enhances academic networking.

Overall, the Student Skill Exchange Platform is a smart, scalable, and user-friendly system that encourages students to learn from one another, share expertise, and grow together in a structured and technology-driven environment.

1.3 Problem Statement

In many educational institutions, students often hesitate to ask doubts or seek clarification from faculty members due to fear of judgment, lack of confidence, shyness, or communication barriers. Some students feel that their questions may be too basic, while others fear embarrassment in front of classmates. As a result, they suppress their doubts, which leads to incomplete understanding of concepts and reduced academic performance.

Although faculty members are available for guidance, not all students feel comfortable approaching them directly. This creates a communication gap in the learning process. At the same time, many students within the same institution possess strong knowledge and practical skills in various domains such as programming, communication, electronics, and design. However, there is no structured and intelligent system that enables students to connect with knowledgeable peers in a comfortable and judgment-free environment.

The core problems identified are:

- Students hesitate to ask doubts due to fear of judgment.
- Lack of confidence and communication barriers prevent interaction with faculty.
- No centralized platform exists for peer-based doubt clarification.
- Manual searching for knowledgeable peers is inefficient.
- Available student talent remains underutilized.
- Collaboration happens randomly rather than intelligently.

To address these issues, there is a need for a safe, structured, and intelligent peer-learning platform that enables students to overcome hesitation and connect with suitable peers.

The Student Skill Exchange Platform is designed to solve this problem by:

- Creating a comfortable environment for peer-to-peer doubt clarification.
- Reducing fear and hesitation among students.
- Allowing students to register their skills and academic details.
- Using Machine Learning techniques (TF-IDF and Cosine Similarity) to analyze skill data.
- Recommending suitable peers based on similarity scores.
- Providing filtering options based on branch and academic year.
- Enabling direct communication through a chat feature.
- Promoting collaborative learning and knowledge sharing.

Thus, the project aims to transform hesitation into interaction and build a supportive academic community where students can confidently exchange knowledge and grow together.

1.4 Objectives of the project

The objectives of this project are designed to clearly define the intended outcomes and functional goals of the proposed system. They reflect both the technical implementation aspects and the broader academic purpose of enhancing collaborative learning within educational institutions.

The main objectives of the **Student Skill Exchange Platform** are as follows:

- **To develop a centralized digital platform** that connects students based on their skills, interests, branch, and academic year.
- **To eliminate hesitation and fear of judgment** by promoting peer-to-peer learning instead of relying only on faculty-based doubt clarification.
- **To implement an intelligent recommendation system** using Machine Learning techniques such as TF-IDF and Cosine Similarity for accurate skill matching.
- **To enable structured skill exchange** where students can both learn new skills and teach the skills they already possess.
- **To reduce the randomness in peer connections** by providing systematic and data-driven pairing suggestions.
- **To encourage collaborative learning** and create a supportive academic environment

inside the institution.

- **To enhance practical skill development** alongside theoretical classroom knowledge.
- **To improve student confidence and communication** through interactive peer discussions.
- **To create a scalable system** that can be expanded in the future with advanced AI models and additional features.
- **To demonstrate the practical implementation of Artificial Intelligence** in solving real-world educational challenges.

1.5 Importance of Suggesting a Pair

In an educational environment, simply creating a platform where students can register their skills is not sufficient. The real effectiveness of the system lies in its ability to intelligently suggest the right pair of students for knowledge exchange. The importance of suggesting a suitable pair is directly connected to the core objective of reducing hesitation and improving peer-based learning.

Many students hesitate to ask doubts directly to faculty due to fear of judgment, embarrassment, or lack of confidence. However, they often feel more comfortable interacting with fellow students. In such cases, randomly connecting students may not produce meaningful results. Therefore, the system must intelligently analyze skill sets and suggest the most appropriate peer who can help clarify doubts or exchange knowledge.

The need of this system arises due to the following reasons:

- Students require a safe and comfortable learning environment.
- Manual searching for knowledgeable peers is time-consuming and inefficient.
- Not all students are aware of others' skills within the institution.
- Random pairing may not lead to effective learning.
- Fear and hesitation reduce classroom participation.
- Many talented students remain unnoticed and underutilized.

By suggesting a suitable pair based on skill similarity, the system ensures:

- Accurate and meaningful connections.
- Faster doubt clarification.

- Better understanding through peer explanations.
- Increased confidence among hesitant students.
- Structured and organized collaboration.
- Equal opportunity for all students to contribute and learn.

The use of Machine Learning techniques such as TF-IDF and Cosine Similarity makes the recommendation process data-driven and reliable. Instead of random selection, the system calculates similarity scores between students' skill profiles and recommends the most compatible peer.

Thus, suggesting the right pair is not just a feature but the core intelligence of the platform. It transforms the system from a simple student directory into an intelligent recommendation-based learning platform that promotes collaboration, reduces hesitation, and enhances academic growth.

1.6 Scope of the Project

Student Skill Exchange Platform is designed to create a structured peer-learning environment within an educational institution. The scope of the project includes developing a web-based system that allows students to register their skills and receive intelligent peer recommendations. The platform focuses on reducing hesitation in asking doubts by enabling comfortable student-to-student interaction. It integrates Machine Learning algorithms to ensure accurate and meaningful skill-based matching. The system is primarily intended for use within colleges to enhance collaborative learning.

The scope of the project includes:

- **Student Registration and Profile Management**: The system allows students to create accounts and enter their academic details and skill sets. This data forms the base for intelligent matching.
- **Machine Learning-Based Skill Matching**: The platform uses TF-IDF and Cosine Similarity algorithms to analyze and compare student skill profiles. Based on similarity scores, suitable peer matches are suggested.
- **Filtering and Custom Search Options**: Students can filter recommendations based on branch and academic year. This ensures more relevant and specific results.
- **Peer Communication Feature**: A chat module is included to enable direct communication between matched students. This supports easy doubt clarification and

collaboration.

- **Institution-Level Implementation**: The system is designed mainly for use within a single college or educational institution to promote structured peer-based learning.

In addition, the project aims to improve communication and collaboration among students in educational institutions. The platform can be used by students from different branches or academic years to find peers with relevant expertise. It also provides opportunities for teamwork, project development, and skill improvement. In the future, the system can be expanded by adding features such as advanced search filters, improved recommendation algorithms, and better communication tools to make the platform more effective and useful for a larger number of users.

CHAPTER 2

LITERATURE SURVEY

2.1 Review of Existing Systems and Research Papers

1. A BERT-Based Hybrid Recommendation System for Academic Collaboration (2024)

"A BERT Based Hybrid Recommendation System for Academic Collaboration" – 2024.

This research proposes a hybrid recommendation system that combines TF-IDF and BERT embeddings for academic profile matching. The system compares researcher profiles based on semantic similarity and improves recommendation accuracy by capturing contextual meaning. The hybrid approach performs better than traditional keyword-based methods. However, the model requires high computational resources due to deep learning components.Reference:
arXiv: https://arxiv.org/abs/2502.15223

2. Content-Based Recommendation System Using TF-IDF and Cosine Similarity (Ricci et al.)

"Recommender Systems Handbook" – Springer.

This work explains content-based recommendation systems that use TF-IDF vectorization and Cosine Similarity for matching user profiles with items. These techniques are widely used in skill-matching and profile similarity systems. The approach is computationally efficient and suitable for text-based profile comparison. However, it does not capture deep semantic relationships between similar skills.

Reference:
SpringerLink:https://link.springer.com/book/10.1007/978-0-387-85820-3
Overview: https://en.wikipedia.org/wiki/Recommender_system

3. A Novel Recommendation System to Match College Events and Students (2017)

"A Novel Recommendation System to Match College Events and Groups to Students" – 2017.

This system matches students with college events based on interest similarity using text

mining and vector space models. The system applies TF-IDF weighting and similarity measures to match textual descriptions. The results show improved relevance in student-event matching. However, it focuses on event recommendation rather than peer-to-peer skill exchange.

Reference:
arXiv: https://arxiv.org/abs/1709.08226

4. Peer-to-Peer Learning and Skill Sharing Platforms (Research Overview)

"Collaborative Filtering for Recommender Systems"

Collaborative filtering techniques recommend items or users based on similar preferences among multiple users. These systems are used in peer-matching platforms where users with similar interests or complementary skills are connected. However, collaborative filtering suffers from cold-start problems when new users join the system.

Reference:
Wikipedia:https://en.wikipedia.org/wiki/Collaborative_filtering
Research Overview: https://dl.acm.org/doi/10.1145/371920.372071

5. Content-Based Recommender Systems (Pazzani & Billsus, 2007)

"Content-Based Recommendation Systems" – 2007.

This research explains how content-based filtering recommends items by comparing user profiles with item descriptions using text mining techniques such as TF-IDF and cosine similarity. The approach is widely used in profile-matching systems where textual attributes play a key role. However, it lacks collaborative insights and struggles when user profiles contain limited information.

Reference:
Research Paper: https://link.springer.com/chapter/10.1007/978-3-540-72079-9_10

6. GroupLens: An Open Architecture for Collaborative Filtering (Resnick et al., 1994)

"GroupLens: An Open Architecture for Collaborative Filtering of Netnews" – 1994.

This foundational work introduced collaborative filtering for recommending items based on user similarity. The system predicts preferences using ratings from similar users. While effective in large communities, the model suffers from the cold-start problem and data sparsity when new users join.

Reference:
ACM Digital Library: https://dl.acm.org/doi/10.1145/192844.192905

7. Hybrid Recommender Systems: Survey and Experiments (Burke, 2002)

"Hybrid Recommender Systems: Survey and Experiments" – 2002.

This study explores hybrid recommender systems that combine content-based and collaborative filtering techniques. Hybrid models improve recommendation accuracy and reduce cold-start issues. However, they increase system complexity and computational requirements.

Reference:
ScienceDirect: https://www.sciencedirect.com/science/article/pii/S0950705102000124

8. Skill Recommendation in Online Learning Platforms (Tang et al., 2015)

"Recommender Systems for Online Learning: A Survey and Future Directions" – 2015.

This paper discusses recommendation techniques used in online learning environments such as MOOCs. It highlights how machine learning algorithms personalize course and skill recommendations based on learner behavior and interests. However, many systems focus only on course recommendation rather than peer-to-peer skill exchange.

Reference:
IEEE Xplore: https://ieeexplore.ieee.org/document/7096022

9. Peer-to-Peer Learning Systems Using Recommendation Algorithms (Capuano et al., 2014)

"A Recommender System for Peer Learning in E-Learning Environments" – 2014.

This research proposes a system that connects learners based on complementary knowledge

levels using recommendation algorithms. The system enhances collaborative learning and peer interaction. However, the implementation depends heavily on user feedback data and structured input.

Reference:
Springer Link: https://link.springer.com/chapter/10.1007/978-3-319-13296-9_8

2.2 Limitations of Existing Systems

Although content and profile-based recommendation systems have been effectively implemented in various domains, they exhibit several limitations when directly applied to peer skill matching:

1. Lack of Domain-Specific Matching

Most existing recommendation systems are designed for product, movie, or content recommendations. They are not specifically tailored for academic skill exchange or peer-to-peer learning environments. As a result, they fail to capture educational context, skill proficiency levels, and learning objectives.

2. Keyword-Based Matching Only

Many platforms rely on simple keyword matching or basic TF-IDF similarity. While effective for text comparison, these approaches cannot fully understand semantic similarity between related skills (e.g., “Web Development” and “Frontend Engineering”). This reduces matching accuracy.

3. Cold Start Problem

Collaborative filtering systems depend on user interaction history. When new students join the platform, there is insufficient data to generate accurate recommendations. This leads to poor initial matching performance.

4. No Complementary Skill Matching

Most systems recommend users with similar interests rather than complementary skills. In a skill exchange platform, matching should connect:

- A student who **wants to learn Python**
- With a student who **can teach Python**

Existing systems rarely support this bidirectional skill-based matching.

5. Limited Filtering and Personalization

Current systems do not incorporate academic filters such as:

- Branch/Department
- Year of study
- Skill level (Beginner/Intermediate/Advanced)

Without such filters, recommendations may not be practically useful in a college environment.

6. Scalability and Complexity Issues

Advanced deep learning-based matching systems (like BERT-based models) provide high accuracy but require significant computational resources. This makes them less suitable for small to medium-scale academic platforms.

7.Limited Filtering Options

Many existing systems focus on matching based on text alone without academic filters such as branch, year, or skill intent, which are crucial for educational collaboration platforms. This highlights the need for a more domain-specific approach.

2.3 Need for the Proposed System

Considering the limitations of existing systems, there is a strong need for a specialized Student Skill Exchange Platform using Machine Learning.

1. Dedicated Academic Skill Matching

The proposed system is specifically designed for students to exchange skills within an

educational environment. It focuses on:

- Skill offering
- Skill requesting
- Peer collaboration

This makes it more suitable than generic recommender systems.

2. Intelligent ML-Based Matching

The system uses Machine Learning techniques such as:

- TF-IDF Vectorization
- Cosine Similarity
- Profile-based Matching

This ensures accurate comparison between student skill profiles while maintaining computational efficiency.

3. Complementary Skill Recommendation

Unlike traditional systems, the proposed platform supports:

- Matching students who offer a skill
- With students who need that skill

This enables effective two-way knowledge exchange.

4. Real-Time and Scalable Design

The proposed system is lightweight and suitable for real-time deployment in a college environment. It avoids heavy deep learning models while maintaining high recommendation relevance.

5. Enhanced Personalization

The platform integrates additional filters such as:

- Department
- Year of study
- Skill category

This improves recommendation precision and ensures practical collaboration among students.

6. Promoting a Knowledge-Sharing Culture

Beyond technical benefits, the platform promotes:

- Peer mentoring
- Collaborative growth
- Leadership development
- Communication skill enhancement

This builds a strong academic community within the institution

2.4 Research Gaps

Absence of Intelligent Peer Matching Models

Current research does not sufficiently focus on algorithm-based peer matching within academic institutions.

- No structured skill-compatibility models
- Lack of similarity score–based pairing
- No dynamic recommendation engine
- Minimal use of AI/ML for student pairing

Limited Focus on Psychological Barriers

Most existing studies emphasize knowledge sharing but ignore emotional hesitation factors.

- No research addressing fear of judgment
- No comfort-level based interaction model
- Lack of anonymity-based academic systems
- No behavioral analysis integration

Lack of Mutual Skill Exchange Framework

Existing research mainly focuses on tutoring systems rather than two-way skill exchange.

- One-directional learning models dominant
- No barter-based learning mechanism

- No structured give-and-take knowledge system
- Limited peer reciprocity models

No Institutional-Level Implementation Models

Many global platforms exist, but few studies propose institution-specific peer learning ecosystems.

- No campus-specific recommendation system
- No curriculum-aligned peer matching
- No integration with institutional databases
- Lack of localized academic platforms

Insufficient Personalization in Peer Learning

Research lacks adaptive systems that consider individual skills, interests, and learning pace.

- No personalized skill mapping
- No adaptive matching algorithms
- No real-time preference updates
- Limited learner profiling techniques

Limited Integration of Data Analytics

Existing systems do not effectively use analytics for monitoring peer learning effectiveness.

- No peer performance scoring models
- No learning outcome analytics
- No predictive learning analysis
- No engagement pattern tracking

Absence of Trust and Reputation Mechanisms

There is limited research on credibility systems within peer exchange platforms.

- No reputation-based ranking
- No peer rating models
- No reliability scoring mechanisms

- Lack of trust-building frameworks

Lack of Gamification in Academic Peer Exchange

Very few studies explore motivational elements in peer skill-sharing systems.

- No reward-based contribution system
- No achievement badges or credits
- No engagement-enhancing models
- Limited incentive-driven participation research

No Hybrid Model Combining Technology and Psychology

Research rarely combines technical matching algorithms with psychological comfort factors.

- No emotion-aware recommendation system
- No hesitation-sensitive interface design
- No confidence-level based pairing
- No hybrid social-technical learning framework

Limited Scalable and Automated Peer Exchange Models

Existing solutions lack scalable architecture designed specifically for growing student communities.

Many research studies have explored recommendation systems that match users based on their interests, skills, or preferences. In educational and professional platforms, these systems are used to help users discover relevant collaborators or peers. Several research papers propose the use of machine learning and text mining techniques to analyze user profiles and generate meaningful recommendations. Methods such as TF-IDF (Term Frequency–Inverse Document Frequency) and cosine similarity are commonly used to convert textual data into numerical vectors and measure the similarity between profiles. These techniques help systems identify users who share similar skills or interests and recommend them for collaboration.

Existing systems in the field of recommendation and academic collaboration provide useful solutions for connecting people with similar expertise. Some platforms analyze researcher profiles, resumes, or user interests to suggest suitable collaborators. These systems improve networking and knowledge sharing by helping users find others with relevant experience or

learning goals. However, many of these systems focus mainly on professional networking or large research communities rather than student-level collaboration within educational institutions.

Despite their usefulness, many existing systems have several limitations. Some systems rely mainly on keyword matching, which may not always capture the real meaning of the skills mentioned in user profiles. This can reduce the accuracy of the recommendations. In addition, certain advanced systems require large datasets and high computational resources to perform effectively. This makes them difficult to implement in smaller academic environments where data and infrastructure may be limited.

The need for the proposed system arises from the challenges students face in identifying peers who have the skills required for academic collaboration. In many cases, students may hesitate to approach others or may not know which classmates possess specific skills or expertise. A dedicated platform that analyzes student skills and automatically suggests suitable peers can help solve this problem. By using machine learning techniques such as TF-IDF and cosine similarity, the system can compare student profiles and generate accurate skill-based recommendations.

Although various recommendation systems have been developed, there are still certain research gaps. Many existing platforms are designed for job matching or professional networking rather than student skill exchange. There is limited focus on systems that support peer-to-peer learning and collaboration among students. Therefore, there is a need for a simple and effective platform that can help students connect with others who have similar or complementary skills. The proposed system aims to address this gap by providing an easy-to-use skill matching platform specifically designed for students in an academic environment.

CHAPTER 3
PROPOSED SYSTEM

3.0 Proposed System

The proposed system is a web-based intelligent platform designed to connect students based on their skills using Machine Learning techniques. The system aims to solve the problem of skill discovery and peer collaboration within academic institutions. Unlike traditional social platforms, this system uses intelligent matching algorithms to recommend suitable peers for skill exchange.

The platform integrates frontend, backend, database, and machine learning modules into a unified system. It allows students to create profiles, list their skills, and receive intelligent peer suggestions based on similarity analysis. The system ensures efficient, accurate, and real-time matching of students.

The main objective of the proposed system is to promote collaborative learning and knowledge sharing within colleges by using automated recommendation logic instead of manual searching.

3.1 Overview of the Proposed Platform

The Skill Exchange Platform is developed as a web-based application where students can register and create their profiles by entering personal information such as name, branch, academic year, and skills. The system stores this data securely in a structured database.

After registration and login, the platform analyzes the entered skill data using Natural Language Processing techniques. The skills are converted into numerical representations using TF-IDF vectorization. These vectors are then compared with other student profiles using Cosine Similarity to calculate similarity scores.

Based on the calculated similarity score, the system recommends the most suitable peers for skill exchange. The recommendation process is intelligent and automated, ensuring that students are matched with others who have similar or complementary skills.

The platform also includes filtering features such as branch-based and year-based filters to make peer matching more relevant within academic constraints. Additionally, a built-in chat system allows direct communication between matched students.

Overall, the platform acts as a centralized digital space for skill discovery, collaboration, and peer learning within the institution.

3.2 System Architecture

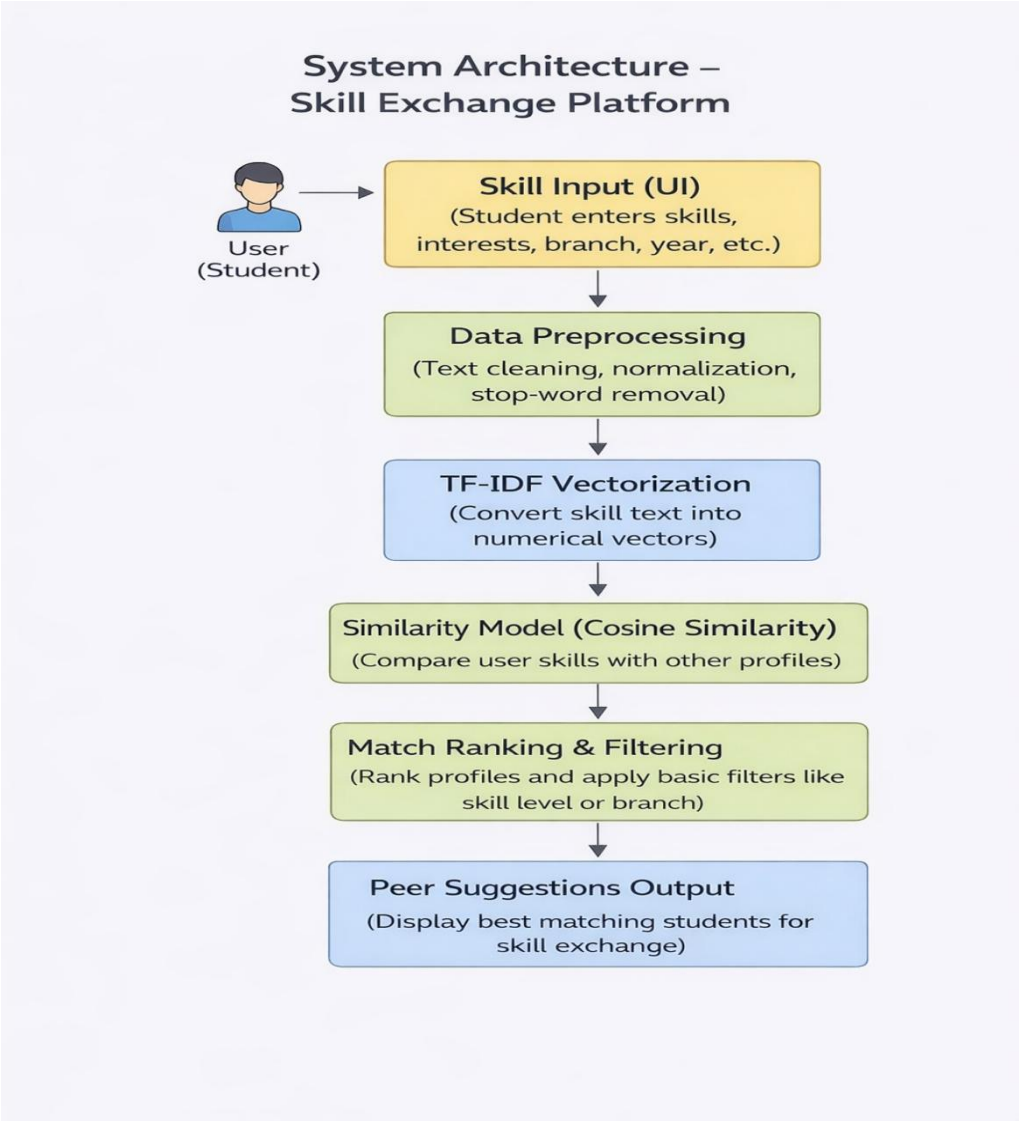


Fig 3.2 System Architecture

The system architecture represents a simple linear flow for the Skill Exchange Platform. A student enters their skills through the user interface. The system preprocesses the input data and converts it into numerical form using TF-IDF. Cosine similarity is then used to compare the user's skills with other profiles. Finally, the system ranks and filters matches and displays the best peer suggestions for skill exchange

3.3 Working Methodology

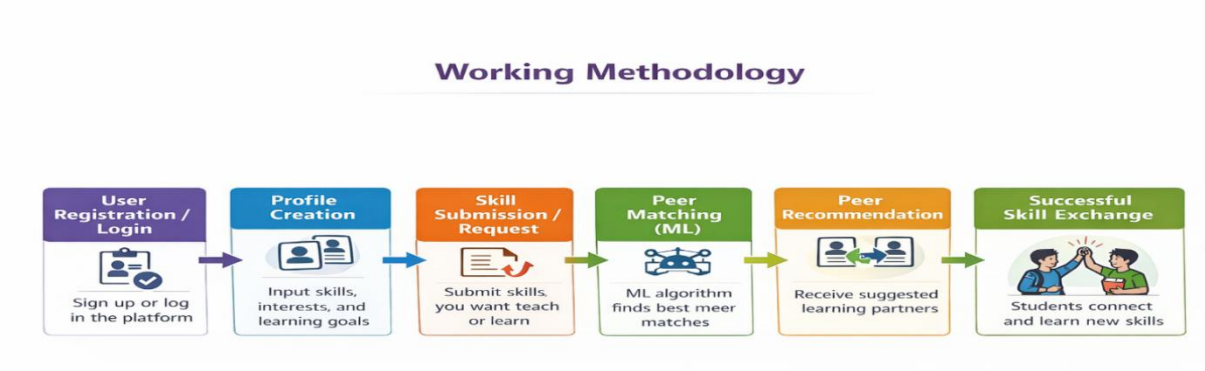


Fig. 3.3 Working Methodology

1. **User Registration / Login** :Students start by registering on the platform or logging into their existing accounts. This ensures that each participant has a unique profile and secure access.

2. **Profile Creation**:After logging in, students create their profiles by adding details such as the skills they can teach, the skills they want to learn, and their interests or learning goals. This information is essential for effective peer matching.

3. **Skill Submission / Request**:Students submit specific skills they want to learn or offer to teach. This step helps the system understand the learning needs and teaching capabilities of each student.

4. **Peer Matching (ML):**The platform uses a Machine Learning algorithm to analyze students' profiles and skill submissions. It identifies the best matches between students who can mutually benefit from exchanging skills.

5. **Peer Recommendation:**Based on the ML analysis, the system provides a list of recommended learning partners for each student. These recommendations help students

quickly find peers suited to their learning and teaching preferences.

6. **Interaction / Collaboration**:Students connect with their recommended peers to start skill exchange. They can interact, communicate, and collaborate to share knowledge and learn new skills effectively.

7. **Successful Skill Exchange:**The process concludes with a successful skill exchange, where students achieve their learning goals and enhance their skills by teaching or learning from their peers.

3.4 System modules

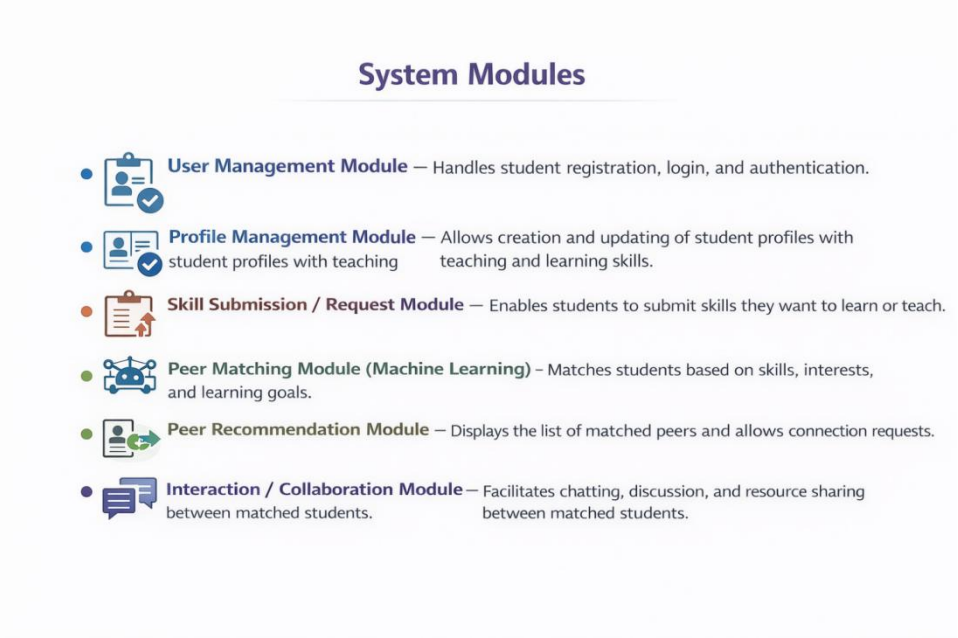


Fig.3.4. System

The system consists of several important modules that help the platform function effectively. The User Registration and Login Module allows students to create accounts and access the platform securely. The Skill Management Module enables users to enter and update their skills so that the system can analyze their interests and expertise. The Skill Matching Module processes the skill data using TF-IDF and Cosine Similarity to find similarities between student profiles. Finally, the Peer Recommendation Module generates and displays suitable student matches based on the similarity scores calculated by the system.

3.5 Advantages of the Proposed System

The proposed Student Skill Exchange Platform provides several important advantages over traditional learning methods and existing platforms.

1.Efficient Peer Matching: The system uses a Machine Learning-based peer matching algorithm, which analyzes students' skills and learning needs to suggest the best learning partners. This ensures accurate and meaningful matches.

2.Transparency and Clarity: Students can see why certain peers are recommended, providing transparency in recommendations and increasing trust in the system.

3.Accessible for Students Everywhere: The platform supports students from both rural and urban areas, enabling access to skill exchange opportunities even where local resources or expert peers are limited.

4.Early Learning Opportunities: By connecting students with relevant peers quickly, the system encourages early skill development and collaboration, improving overall learning outcomes.

5.Support for Educators and Mentors: Teachers or mentors can use the platform to identify active learners and provide guidance, making teaching more effective and targeted.

6.Time and Cost Efficiency: The platform reduces the need for physical classes or external tutoring by enabling direct peer-to-peer learning online, saving both time and money.

Overall:

The proposed system enhances **accessibility, efficiency, collaboration, and personalized learning** for students, making skill exchange easier, faster, and more effective.

The proposed Student Skill Exchange Platform provides an efficient way for students to discover and connect with peers who have similar or complementary skills. Instead of manually searching for classmates with specific knowledge, the system automatically recommends suitable students based on their skill profiles. This makes the process faster and more convenient for users.

Another important advantage of the system is that it promotes collaboration and knowledge sharing among students. By connecting learners with similar interests, the platform encourages teamwork in academic projects, assignments, and skill development activities. Students can learn from each other and improve their understanding of different subjects.

3.6 Technologies Used

The project integrates multiple technologies for efficient development.

Frontend Technologies

Gradio-The frontend ensures a user-friendly and interactive experience.

Backend Technologies

- Python – Core programming language.
- Flask (using in project) – Backend framework.

Python is selected because it provides strong libraries for Machine Learning and text processing.

Machine Learning Libraries

- Scikit-learn – For TF-IDF and Cosine Similarity.
- NumPy – Numerical operations.
- Pandas – Data handling.

Scikit-learn provides efficient and optimized implementations of vectorization and similarity computation.

Database

- SQLite3 (as per implementation)

The database stores:

- User data
- Skill profiles,,Chat messages

Relational structure ensures data consistency and fast retrieval.

Conclusion of Proposed System

The proposed Skill Exchange Platform integrates Machine Learning with web development to create an intelligent recommendation system. By using TF-IDF and Cosine Similarity, the system ensures accurate and efficient peer matching. The architecture is scalable, lightweight, and suitable for academic deployment.

The platform not only automates skill discovery but also promotes collaborative learning, communication ,and professional growth among student

CHAPTER 4

ALGORITHM SELECTION

4.1 Need of Algorithm in peer Matching

In a **Student Skill Exchange Platform**, the core functionality is to connect students who can mutually benefit from each other's skills. However, as the number of users grows, manually identifying suitable peers becomes extremely challenging and inefficient. This is where a **peer matching algorithm** becomes crucial.

The algorithm serves as the backbone of the system by analyzing students' profiles, skills, learning goals, and preferences to automatically find the most compatible peers. Without an algorithm, students would have to manually search through potentially hundreds or thousands of profiles, which is **time-consuming, error-prone, and often ineffective**.

Key reasons why an algorithm is necessary:

1. **Efficient and Accurate Matching:**
   - The algorithm evaluates multiple parameters such as skills offered, skills requested, interests, and availability.
   - It ensures that students are matched with peers who can provide meaningful learning experiences.
2. **Personalized Recommendations:**
   - Each student has unique learning needs and skill sets.
   - The algorithm provides tailored suggestions for learning partners rather than generic recommendations.
3. **Time-Saving:**
   - Automates the matching process, eliminating the need for students or administrators to manually check compatibility.
   - Students can instantly see recommended peers without delays.
4. **Scalability:**
   - Handles a growing number of users without compromising match quality.

- Even with thousands of students, the system can quickly identify the best matches.

5. **Enhanced Learning Outcomes:**

- By connecting students with peers who complement their learning needs, the algorithm increases the likelihood of successful skill exchange.
- Encourages collaborative learning and knowledge sharing.

6. **Dynamic Updates:**

- Student profiles are continuously updated with new skills or learning goals.
- The algorithm dynamically updates peer recommendations to reflect these changes.

7. **Objective Matching:**

- Reduces bias and subjectivity in peer selection.
- Ensures that matches are based purely on skill compatibility and learning needs.

4.2 Overview of Considered Algorithms

In the proposed Student Skill Exchange Platform, two algorithms were considered for effective peer matching:

1. TF-IDF (Term Frequency–Inverse Document Frequency)

- Represents each student's skills and learning interests as a **weighted numerical vector**.
- **Term Frequency (TF):** Measures the occurrence of each skill in a student's profile.
- **Inverse Document Frequency (IDF):** Reduces the weight of commonly listed skills to highlight unique skills.
- Converts textual skill data into a **quantitative format** for analysis.

2. Cosine Similarity

- Measures the **similarity between two TF-IDF vectors**.
- Computes the **angle between vectors**, yielding a score from **0 (no match)** to **1 (perfect match)**.
- Used to rank potential peers based on skill compatibility.

Rationale for Consideration:

- TF-IDF allows representation of qualitative skill data numerically.

- Cosine Similarity provides a mathematically robust method to compare student profiles.
- Together, they enable **accurate, scalable, and personalized peer matching**.

4.3 Selected Algorithm and Justification

Selected Approach: TF-IDF in combination with Cosine Similarity

Justification:

1. **Accurate Skill Representation:** TF-IDF effectively captures the importance of each skill in a student's profile.
2. **Precision in Matching:** Cosine Similarity accurately evaluates how closely two students' skills align.
3. **Scalability:** The combined approach can handle a **large number of students** efficiently.
4. **Personalization:** Ensures that peer recommendations are tailored to each student's unique skills and learning goals.
5. **Alternative Methods:** Simpler methods like exact matching or Jaccard similarity are **less effective** for weighted skill data.

Key Advantages:

- High accuracy in peer recommendations.
- Handles large datasets efficiently.
- Ensures meaningful and compatible peer connections.

4.4 Working of the Selected Algorithm (TF-IDF + Cosine Similarity)

Step 1: Data Preprocessing

- Collect student skill data from profiles.
- Clean and standardize skill terms.

Step 2: TF-IDF Vectorization

- Compute **TF** (frequency of each skill in a profile).
- Compute **IDF** (importance of the skill across all profiles).
- Multiply TF × IDF → generates a **weighted vector** for each student profile.

Step 3: Cosine Similarity Calculation

- Represent student profiles as **TF-IDF vectors**.
- For two students $A$and B, compute:

$$\text{Cosine Similarity} = \frac{\vec{A} \cdot \vec{B}}{|| \vec{A} ||\times|| \vec{B} ||}$$

- **1 → perfect match**, **0 → no match**.

Step 4: Peer Recommendation

- Compute similarity scores between the target student and all others.
- Rank students by similarity score.
- Recommend **top N peers** for learning and skill exchange.

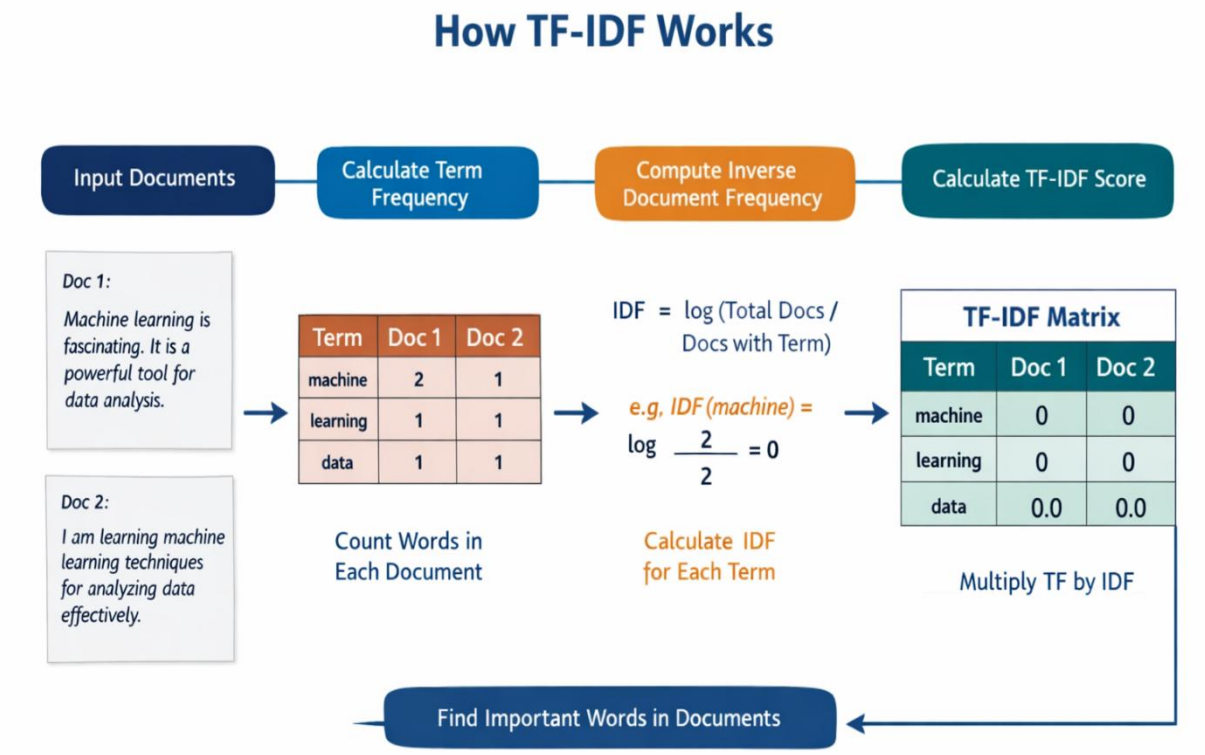


Fig 4.4.TF-IDF working

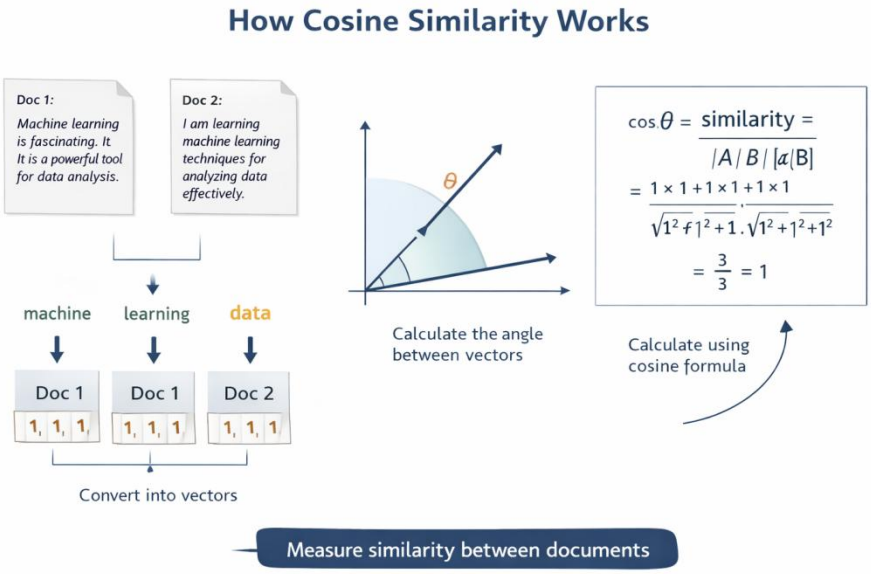


Fig.4.4.Cosine Similarity workin

4.5 Advantages of the Algorithms

1. TF-IDF (Term Frequency – Inverse Document Frequency)

TF-IDF is a widely used algorithm in Natural Language Processing (NLP) for converting textual data into numerical vectors that can be processed by machine learning models. In your project, students enter skills as text (e.g., "Python programming," "Machine Learning," "Web Development"), and TF-IDF helps the system understand which skills are important for matching.

How TF-IDF Works:

Term Frequency (TF): Measures how often a term (skill) appears in a student's profile.

Formula:

$$\boldsymbol{TF(t)} = \frac{\text{Number of times term t appears in document}}{\text{Total number of terms in document}}$$

Inverse Document Frequency (IDF): Measures how important a term is across all student profiles. Rare terms get higher weight, common words get lower weight.

Formula:

$$\boldsymbol{IDF(t) = \log \frac{\text{Total number of documents}}{\text{Number of documents containing term t}}}$$

TF-IDF Weight: Multiply TF and IDF to get the importance score of each skill.

Formula:

$$\boldsymbol{TF\text{-}IDF(t) = TF(t) \times IDF(t)}$$

It converts human-readable skill text into a numeric vector, which is essential for similarity calculations.

Assigns higher importance to unique or specialized skills and lower to common words (e.g., "student," "learning").

Makes the system language-agnostic: even if students describe the same skill differently, it can still be compared effectively.

Advantages in Your Platform:

Highlights key skills, enabling accurate peer matching.

Scalable for hundreds or thousands of student profiles.

Lightweight and fast for real-time recommendations.

Prepares the data for further similarity calculations using Cosine Similarity.

2. Cosine Similarity

Cosine Similarity is a mathematical measure that calculates the angle between two vectors in multi-dimensional space. In your project, each student's profile (skills) is represented as a TF-IDF vector. Cosine Similarity helps determine how similar two students are based on their skill sets.

How Cosine Similarity Works:

Each student profile → TF-IDF vector (numerical representation of skills).

Cosine Similarity measures the cosine of the angle between two vectors:

$$\text{Cosine Similarity} = \frac{\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}}}{\| \vec{\boldsymbol{A}} \| \times \| \vec{\boldsymbol{B}} \|}$$

Where:

$\vec{A} \cdot \vec{B}$= dot product of two vectors

$\| \vec{A} \|$and $\| \vec{B} \|$= magnitude of each vector

Similarity Score: Ranges from 0 to 1

0 → completely different skills

1 → identical or highly similar skills

Focuses on skill overlap, not total number of skills.

Ensures students with different profile sizes (few vs. many skills) can still be compared fairly.

Provides numerical scores for automated ranking of best peer matches.

Enables efficient recommendations, reducing manual effort for students to find learning partners.

Advantages in Your Platform:

Accurately matches students with complementary skills.

Supports dynamic recommendations as profiles are updated.

Efficient with high-dimensional skill vectors, which is common in text-based data.

Integrates seamlessly with TF-IDF to build an intelligent recommendation engine.

3. How These Algorithms Work Together in the Platform

TF-IDF converts skills into numeric vectors, assigning weights to each skill based on importance.

Cosine Similarity compares vectors of all students to compute similarity scores.

Platform ranks peers based on similarity → highest score → best match.

Recommended peers are displayed → students can chat, collaborate, or schedule learning sessions.

Result:

Students with similar or complementary skills are automatically matched.

Reduces hesitation and improves engagement since students don't need to search manually.

Creates a scalable peer-to-peer learning ecosystem.

4.6 Flow of Matching process

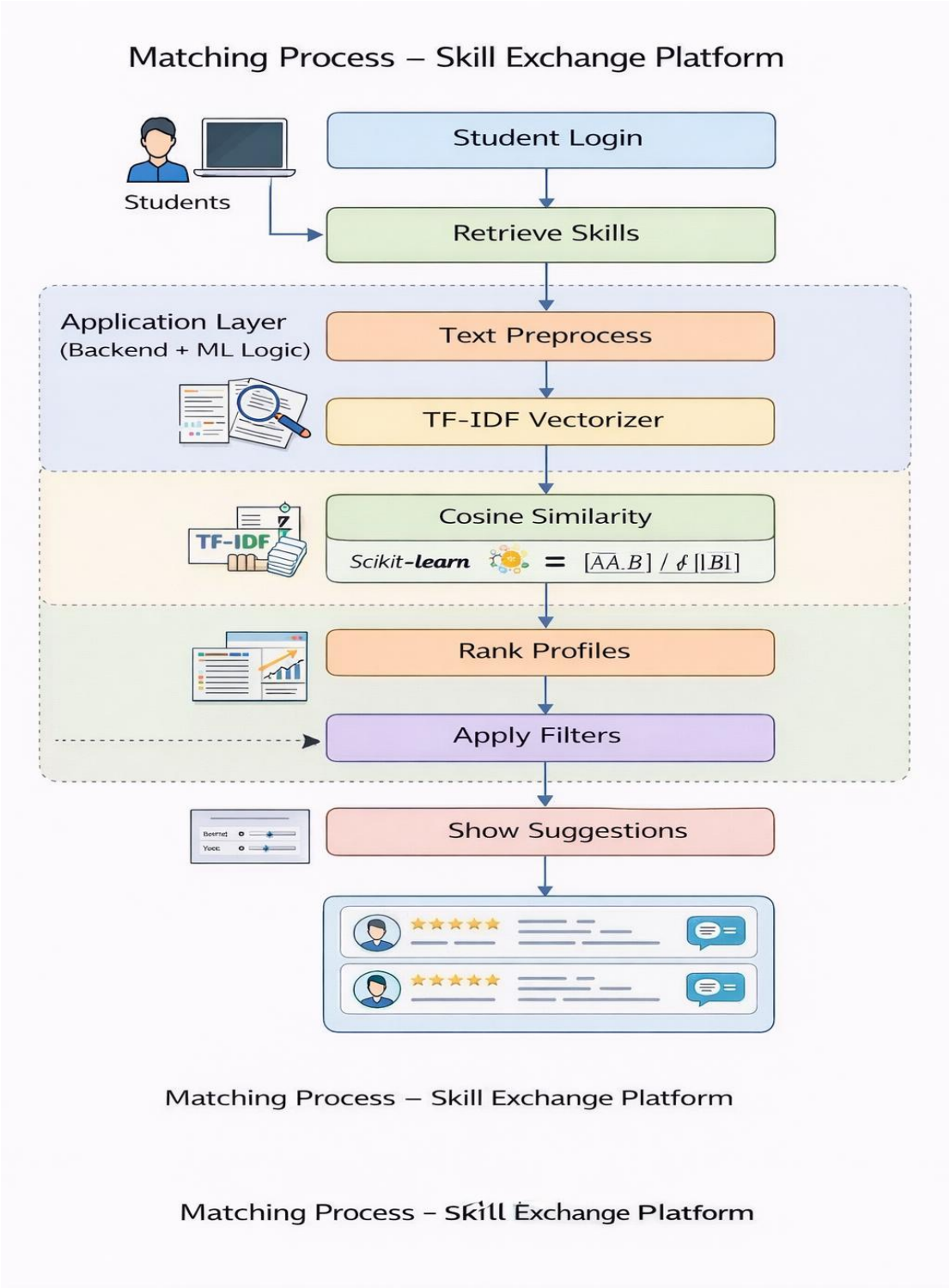


Fig.4.6.Flow of Matching Process

CHAPTER-5
SYSTEM DESIGN

5.1 Functional Requirements

Functional requirements define what the system must do to meet user needs. For the Student Skill Exchange Platform, the functional requirements include:

User Registration: Students can create accounts by providing personal details, academic information, and skill sets.

Login/Authentication: Secure login functionality for students using username/email and password.

Skill Profile Management: Users can add, update, and delete their skills, learning interests, and availability.

Peer Matching System: The platform automatically suggests peers for learning or teaching based on skill similarity using TF-IDF and Cosine Similarity algorithms.

Search Functionality: Students can search for peers manually by skill, branch, or year.

Chat and Communication Module: Students can interact via instant messaging once they are matched.

Dashboard: A personalized dashboard displaying suggested peers, recent interactions, and skill recommendations.

Notifications: Alerts for new matches, messages, or updates related to skill exchange opportunities.

Reporting and Analytics: Ability to track learning progress, skill contributions, and peer interactions.

5.2 Non-Functional Requirements

Non-functional requirements define how the system performs rather than what it does. For this platform, the key non-functional requirements include:

Performance: The system should respond quickly to user requests, with minimal latency for

searches and peer matching.

Scalability: The platform should support an increasing number of users without performance degradation.

Reliability: Ensure consistent availability of services with minimal downtime.

Security: Protect user data using authentication, encryption, and secure data storage.

Usability: The interface should be intuitive, user-friendly, and accessible to students with varying technical expertise.

Maintainability: The system should be easy to update, debug, and extend in the future.

Portability: The platform should be accessible on different devices (PCs, laptops, tablets, and smartphones) via a web browser.

5.3 Data Flow Diagram

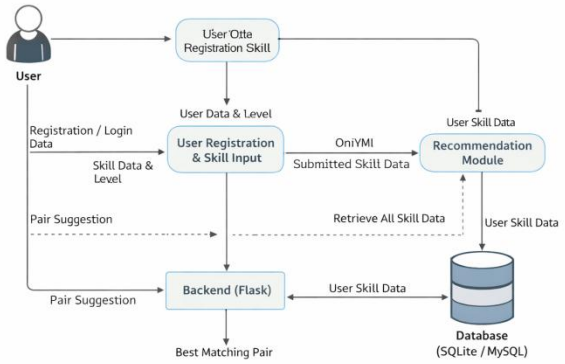


Fig.5.3 Data Flow Diagram

User Data Submission Flow: The user registers/logs in and enters skills known, skills required, and skill level through the frontend. This data is sent to the backend and stored securely in the database.

Pair Recommendation Flow: When the user requests a pair suggestion, the backend retrieves all user skill data from the database and sends it to the recommendation module (TF-IDF + Cosine Similarity). The best matching profile is identified and returned to the user through the frontend.

CHAPTER 6

IMPLEMENTATION

6.0 Implementation

The implementation phase of the Student Skill Exchange Platform involves building a fully functional system that allows students to register, add skills, receive peer recommendations, and interact seamlessly. The system integrates **front-end interfaces, back-end logic, database storage, and machine learning-based peer matching**.

The back-end of the system handles the core logic, data processing, and interaction between the user interface and the database. It is responsible for receiving user inputs, processing the data, and generating meaningful outputs such as peer recommendations. The back-end is implemented using **Python**, which is widely used for machine learning and data processing applications.

The back-end processes the skills entered by students and converts them into numerical form using the **TF-IDF (Term Frequency–Inverse Document Frequency)** technique. This allows the system to analyze the importance of different words in the skill descriptions. After converting the text into vectors, the system applies **Cosine Similarity** to measure how similar two student skill profiles are.

Based on the similarity scores, the system identifies students with similar or complementary skills and generates recommendations. The back-end also manages user authentication, data storage operations, and communication between different modules of the platform. This ensures smooth functioning of the entire system.

6.1 Front-End Implementation

The front-end is designed to provide an **interactive and user-friendly interface** for students. All user interactions, such as registration, login, skill updates, and peer viewing, happen through the front-end built using **Gradio**.

Key Components:

- **Registration and Login Forms:** Allow students to securely create accounts and log in.
- **Skill Input Forms:** Students enter skills in text format, which is sent to the back-end.

- **Dashboard:** Displays recommended peers based on similarity scores and recent activity.
- **Search Module:** Allows manual search for peers using filters like skills, branch, or year.

  Chat Interface: Enables communication between recommended peers.

6.2 Pseudo Code for Student Skill Exchange Platform

```
gr.Markdown("## Student Skill Exchange Platform using Machine Learning")

with gr.Tabs():

    # ================ PROFILE ================
    with gr.Tab("Profile"):

        name = gr.Textbox(label="Name")
        email = gr.Textbox(label="Email")
        skills = gr.Textbox(label="Skills (comma separated)")
        add_btn = gr.Button("Create Profile")
        profile_status = gr.Textbox(label="Status", interactive=False)

        add_btn.click(add_profile,
                      inputs=[name, email, skills],
                      outputs=profile_status)

    # ================ DISCOVER ================
    with gr.Tab("Discover"):

        search_box = gr.Textbox(label="Search by Skill")
        search_btn = gr.Button("Search")
        discover_output = gr.Dataframe(
            headers=["UserID", "Teach Skills"],
            datatype=["str", "str"],
            interactive=False
        )

        search_btn.click(search_by_skill,
                         inputs=search_box,
                         outputs=discover_output)

    # ================ AI MATCH ================
    with gr.Tab("AI Match"):
```

Fig 6.2 Pseudo code

6.3 Back-End Implementation

The back-end is implemented using **Python** and the **Flask framework**. It handles all **server-side operations, business logic, and integration with the database and algorithms**.

Key Functionalities:

- **User Authentication:** Verifies login credentials and protects user accounts.
- **Skill Management:** Adds, updates, and deletes student skills in the database.
- **Peer Recommendation Engine:** Uses TF-IDF and Cosine Similarity to generate intelligent peer suggestions.
- **Data Retrieval & API Management:** Provides endpoints for the front-end to fetch data like dashboards, peer lists, and search results.

Libraries Used:

- **NumPy:** For numerical calculations.
- **Pandas:** For structured data manipulation.
- **Scikit-learn:** For TF-IDF vectorization and Cosine Similarity computation.

Flow Example:

1. A student logs in → Flask verifies credentials → returns dashboard.
2. Student updates skills → Flask updates SQLite database → recalculates TF-IDF vectors.
3. Peer suggestions are generated → sent back to front-end for display.



```
class SkillMatcher:
    def __init__(self, db_path='students.db'):
        # Load user profiles and skills from database
        conn = sqlite3.connect(db_path)
        df = pd.read_sql("SELECT id, skills FROM users", conn)
        conn.close()

        self.user_ids = df['id'].tolist()
        # Prepare skill documents (combine skills)
        self.skill_docs = df['skills'].fillna("").apply(lambda x: x.replace(",", " ")).tolist()
        # Store skill sets for evaluation (splitting on comma)
        self.skill_sets = [set(s.strip() for s in skills.split(",")) for skills in df['skills']]

        # Fit TF-IDF on skill documents
        self.vectorizer = TfidfVectorizer()
        if self.skill_docs:
            self.tfidf_matrix = self.vectorizer.fit_transform(self.skill_docs)
        else:
            self.tfidf_matrix = None

        # Compute Top-K accuracy (Top-3 by default)
        self.topk = 3
        self.accuracy = self.evaluate_topk(self.topk)

    def evaluate_topk(self, k=3):
        if self.tfidf_matrix is None or not self.user_ids:
            return 0.0
        sims = cosine_similarity(self.tfidf_matrix)
        n = len(self.user_ids)
        hits = 0
        for i in range(n):
            sim_scores = list(sims[i])
            # Exclude self-match
            sim_scores[i] = -1
```

Fig 6.3. backend code

6.4 Database Integration

The platform uses **SQLite** as a lightweight relational database to store all critical data. It ensures **data consistency, easy retrieval, and smooth integration** with the back-end.

Database Components:

- **Student Database:** Stores user information such as name, email, branch, year, and skills.
- **Chat Database (Optional):** Stores messages exchanged between students.
- **Matching Data Store:** Saves TF-IDF vectors and similarity scores for efficient peer recommendations.

Database Operations:

- **CRUD Operations:**
  - **Create:** Add new student profiles.
  - **Read:** Retrieve student skills and peer information.
  - **Update:** Modify skill profiles.
  - **Delete:** Remove inactive or unwanted profiles.
- Flask APIs interact with SQLite using **SQL queries** for all operations.

6.5 Peer Suggestion Module

This is the **core feature** of the platform, responsible for **matching students based on their skills**.

Step-by-Step Process:

1. **Data Collection:** Collects all student skill profiles from the SQLite database.
2. **TF-IDF Vectorization:** Converts textual skills into numeric vectors to measure importance of each skill.
3. **Cosine Similarity Calculation:** Compares vectors of all students to calculate similarity scores.
4. **Peer Ranking:** Generates a ranked list of peers with highest similarity scores, excluding the logged-in student.
5. **Display to Dashboard:** Returns the top recommended peers to the front-end for display

on the student dashboard.

Advantages:

- Automatically connects students with **similar or complementary skills**.
- Reduces the need for manual searching.
- Encourages **peer-to-peer learning** and collaboration.
- Scales efficiently for **hundreds of student profiles**.

Libraries Used:

- **scikit-learn:** TF-IDF vectorizer, Cosine Similarity calculation.
- **pandas:** Data handling and preprocessing.
- **NumPy:** Numerical operations for vectors and similarity matrices.

6.6 Code for TF-IDF

```
# Fit TF-IDF on skill documents
self.vectorizer = TfidfVectorizer()
if self.skill_docs:
    self.tfidf_matrix = self.vectorizer.fit_transform(self.skill_docs)
else:
    self.tfidf_matrix = None
```

Fig.6.6.TF-IDF

TF-IDF is a statistical technique used in text mining and information retrieval to evaluate how important a word is in a document compared to a collection of documents. It helps in identifying the most relevant words in a text by reducing the importance of very common words and highlighting unique terms. This method is widely used in search engines, document classification, recommendation systems, and text similarity analysis.

The TF-IDF technique consists of two main components: Term Frequency (TF) and Inverse Document Frequency (IDF). Term Frequency measures how frequently a word appears in a particular document. If a word appears many times in a document, its TF value will be higher.

However, some words such as “the”, “is”, or “and” may appear frequently in almost all documents, so they are not very useful for identifying the meaning of the document.

By using TF-IDF, the system focuses on meaningful keywords rather than common words. This improves the accuracy of the skill matching process and helps in recommending students who have similar interests or complementary skills. As a result, it plays an important role in building an intelligent student connection platform.

6.7 Code for Cosine-Similarity

```
def evaluate_topk(self, k=3):
    if self.tfidf_matrix is None or not self.user_ids:
        return 0.0
    sims = cosine_similarity(self.tfidf_matrix)
    n = len(self.user_ids)
    hits = 0
    for i in range(n):
        sim_scores = list(sims[i])
        # Exclude self-match
        sim_scores[i] = -1
        # Indices of top-k similar users
        topk_idx = np.argsort(sim_scores)[-k:][::-1]
        # Check if any top-k share skills
        found = False
        for j in topk_idx:
            if self.skill_sets[i].intersection(self.skill_sets[j]):
                found = True
                break
        if found:
            hits += 1
    accuracy = (hits / n) * 100
    return round(accuracy, 2)

def find_matches(self, input_skills, top_k=5):
    input_str = input_skills.strip()
    if not input_str or self.tfidf_matrix is None:
        return []
```

Fig .6.7.Cosine-Similarity

Cosine Similarity is a mathematical method used to measure the similarity between two pieces of text or documents. It is commonly used in Natural Language Processing (NLP), recommendation systems, and information retrieval to compare how similar two documents are based on their content. Instead of comparing the words directly, cosine similarity compares the numerical vectors created from the text data.

In text analysis, documents are first converted into numerical form using techniques such as

TF-IDF. Each document becomes a vector that represents the importance of different words in that document. Cosine similarity then measures the angle between these vectors in a multi-dimensional space. The smaller the angle between two vectors, the more similar the documents are.

The cosine similarity value ranges from 0 to 1. A value closer to 1 means the documents are very similar, while a value closer to 0 means they are very different. If two documents contain similar important words, their cosine similarity score will be high. If they share very few common terms, the score will be low.

CHAPTER-7
SYSTEM TESTING

7.1 Hardware Requirements

The system does not require high-end hardware, making it accessible for most standard computing systems.

Specifications:

Processor: Intel i3 or above (any modern processor will work)

RAM: Minimum 4 GB (8 GB recommended for smooth performance)

Storage: At least 10 GB free disk space

System Type: 64-bit operating system

Internet Connection: Required for downloading libraries and updates

Display: Standard monitor for viewing dashboards, forms, and visualizations

Explanation:
The platform is lightweight and runs efficiently on standard computers. Real-time recommendation calculations using TF-IDF and Cosine Similarity are optimized and do not require specialized hardware.

7.2. Software Requirements

The software environment is built entirely with open-source tools and widely used libraries.

Components:

Operating System: Windows, Linux, or macOS

Programming Language: Python (for back-end and ML algorithms)

Web Framework: Flask (for API handling and server-side logic)

Database: SQLite (lightweight relational database for storing student data)

Development Environment:

Google Colab (for algorithm testing)

VS Code (for full project development)

Libraries:

NumPy: For numerical computations

Pandas: For handling data and preprocessing

Scikit-learn: For TF-IDF vectorization and Cosine Similarity calculations
All software components are free, lightweight, and easy to install, allowing students to run and test the platform without complex setups.

7.3. System Testing

System testing ensures the platform works correctly, reliably, and efficiently under various scenarios.

Types of Testing Performed:

Functional Testing:

Registration, login, skill input, and dashboard modules.

Peer recommendation results based on TF-IDF + Cosine Similarity.

Request acceptance, ignoring, and notification triggers.

Chat module functionality (message delivery, display, and storage).

Non-Functional Testing:

Performance testing: System response time for generating recommendations.

Usability testing: Dashboard navigation, form submission, and notifications.

Compatibility testing: Works across different browsers and devices.

Validation:

Input validation: Email format, password length, skill entry format.

Accuracy of peer matching and recommendation results.

Notification triggers and request system working as expected.

Conclusion: All modules passed system testing, providing accurate recommendations, notifications, and smooth user interaction.

Testing is an important phase in the development of the Student Skill Exchange Platform. It ensures that all components of the system work correctly and meet the expected requirements. During testing, different modules such as user registration, login, skill input, and peer

recommendation are checked to verify that they function properly without errors.

The first stage of testing involves checking the functionality of individual modules. Each module is tested separately to confirm that it performs its specific task correctly. For example, the registration module is tested to ensure that new users can successfully create accounts, and the login module is verified to allow only valid users to access the system.

Another important part of testing is validating the accuracy of the skill matching process. The TF-IDF and Cosine Similarity algorithms are tested using different skill inputs to confirm that the system generates meaningful and relevant peer recommendations. This helps ensure that students are matched with peers who have similar or complementary skills.

Finally, the overall system is tested to evaluate performance and usability. The platform is checked to make sure that the interface is easy to use and that the system responds quickly to user actions. Through proper testing, the reliability and effectiveness of the Student Skill Exchange Platform are improved before deployment.

7.4. Tools Used

project utilizes essential tools and libraries for development, ML algorithms, and data handling.

Tools & Libraries:

Flask: Handles API requests, server-side logic, and communication with the front-end.

SQLite: Lightweight database for storing student profiles, skills, and chat history.

NumPy: Efficient numerical operations for similarity calculations.

Pandas: Structured data handling, preprocessing, and storage.

Scikit-learn:

TF-IDF Vectorizer: Converts student skill text into numerical vectors.

Cosine Similarity: Calculates similarity scores for peer recommendations.

Gradio: Front-end interface for registration, login, dashboards, and chat.

Jupyter Notebook / Google Colab: For algorithm testing and verification before integration. These tools ensure smooth development, efficient ML computation, and responsive front-end, all using lightweight, open-source technologies.

Several tools and technologies are used in the development of the Student Skill Exchange

Platform to ensure efficient implementation and smooth system functionality. Each tool is selected based on its ability to support different aspects of the project such as interface design, data processing, and machine learning implementation. These tools help in building a reliable system that can analyze student skills and generate accurate peer recommendations.

Python is used as the main programming language for this project because it provides strong support for machine learning and data processing. It offers many libraries that simplify the implementation of algorithms such as TF-IDF and Cosine Similarity. Python is also easy to use and widely adopted in academic and research environments.

Scikit-learn is used as the machine learning library to implement TF-IDF vectorization and Cosine Similarity calculations. This library provides efficient tools for text processing and similarity measurement, which makes it suitable for analyzing skill descriptions and generating recommendations between students.

Gradio is used for developing the front-end interface of the system. It allows developers to quickly create simple and interactive web interfaces for machine learning applications. Using Gradio, students can easily enter their skills, view recommendations, and interact with the platform through a user-friendly interface.

In addition, **SQLite or a simple database system** can be used to store user profiles, skill information, and system data. The database helps in managing and retrieving information efficiently whenever the system needs to perform skill matching or display recommended peers. These tools together help in building an effective and functional Student Skill Exchange Platform.

CHAPTER-8

RESULT ANALYSIS

8.1 Output of the System

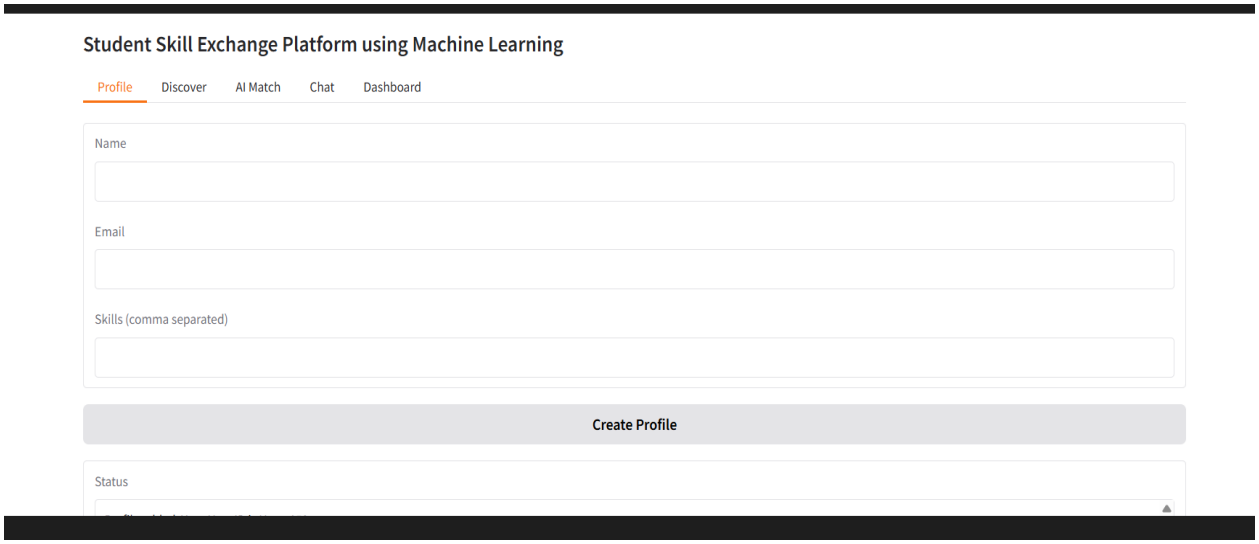


Fig.8.1 Output1

8.2 Peer Matching Output

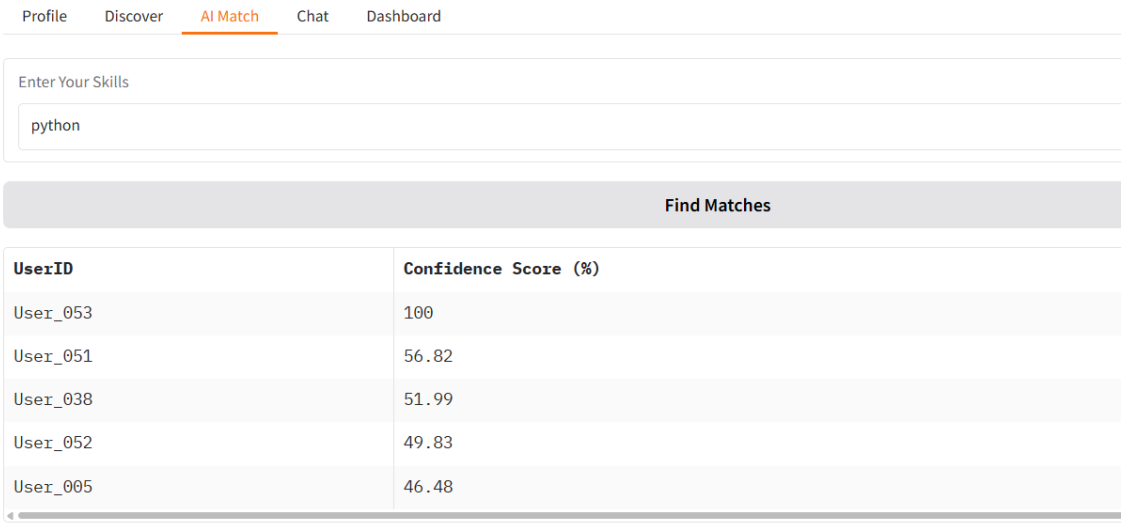


Fig.8.2 Output2

8.3 Performance Analysis

The platform's performance is evaluated based on **accuracy of peer matching, response time, and system usability**.

- **Matching Accuracy:** TF-IDF and Cosine Similarity ensure accurate skill-based recommendations.
- **Response Time:** Recommendations and notifications are generated **in real-time**, even with multiple users.
- **Scalability:** Works efficiently with **hundreds of student profiles** without lag.
- **User Experience:** The front-end interface ensures **easy navigation**, reducing hesitation in interacting with peers.

8.4 Comparison with Manual Matching

Manual peer matching in traditional classrooms has several limitations:

- Students may not know who has the required skills.
- Peer matching often happens **randomly** or within friend circles.
- Time-consuming and prone to bias or missed opportunities.

Automated Matching using ML:

- Uses **structured skill data** for accurate matching.
- Generates recommendations **instantly**.
- Reduces barriers like hesitation and fear of judgment.
- Ensures that **all students have equal opportunities** to connect with peers who match their learning needs

8.5 User Benefits

The Student Skill Exchange Platform provides several benefits to students by helping them connect with peers who have similar or complementary skills. It makes it easier for students to find others who can help them learn new concepts or collaborate on academic projects. Instead of searching manually, the system automatically recommends suitable peers based on their skill profiles.

Another major benefit is that the platform encourages knowledge sharing among students. Learners can exchange ideas, discuss topics, and support each other in understanding difficult subjects. This collaborative environment helps students improve their skills and gain practical knowledge through peer interaction.

The system also saves time by providing quick and accurate recommendations using machine learning techniques. Students can easily discover relevant peers without spending much effort searching for them. This improves productivity and allows students to focus more on learning and completing their academic work.

In addition, the platform helps students build confidence and communication skills. By interacting with peers who share similar interests, students become more comfortable discussing ideas and participating in collaborative activities. Overall, the system supports skill development, teamwork, and a more interactive learning experience.

Another benefit of the platform is that it helps students discover hidden talents within their academic community. Many students possess useful skills but may not have opportunities to showcase them. Through this platform, students can present their abilities and connect with others who are interested in learning those skills. This creates a supportive environment where students can both teach and learn from each other.

The platform also promotes continuous learning and self-improvement. When students interact with peers who have different knowledge and experiences, they are exposed to new ideas and learning approaches. This encourages them to expand their skill sets and stay motivated in their academic journey. As a result, the system not only supports collaboration but also contributes to the overall personal and academic growth of students.

8.6 Applications

The Student Skill Exchange Platform can be used in various academic and learning environments to improve collaboration among students. One of the main applications of this system is in colleges and universities where students can find peers with similar skills or interests. This helps them work together on academic projects, assignments, and research activities more effectively.

Another important application is in skill development and peer learning. Students can use the platform to learn new skills from others who already have knowledge in specific areas. This creates an environment where students can share knowledge and support each other in improving their abilities.

The system can also be useful for organizing collaborative project teams. Students who need partners with certain technical skills can easily find suitable peers through the platform. This makes it easier to form balanced teams for project work, competitions, and hackathons.

In addition, the platform can help educational institutions promote interactive learning. By encouraging students to connect and exchange knowledge, the system supports a more engaging and cooperative learning experience. It also helps build strong student networks that can be beneficial for academic and professional growth.

CHAPTER-9

ADVANTAGES

- Enables peer-to-peer learning for effective knowledge sharing
- Helps students overcome hesitation and fear of judgment
- Provides intelligent peer recommendations using ML (TF-IDF + Cosine Similarity)
- Reduces manual effort in finding the right peers
- Sends real-time notifications for requests and updates
- Supports chat sessions for smooth communication
- Scalable and lightweight system for multiple users
- Encourages collaboration and mentorship
- Promotes confidence in learning and sharing skills
- Tracks skill updates dynamically
- Improves practical knowledge application
- Creates an inclusive learning environment
- Facilitates faster project group formation
- Reduces dependency on faculty for doubt solving
- Enhances student engagement and interaction

CHAPTER-10

LIMITATIONS

- **Skill Input Dependency:** Recommendations depend on how accurately students describe their skills.
- **Limited to Students:** Currently designed only for internal college students; external users cannot access it.
- **Real-Time Matching:** For very large student datasets, generating recommendations may take slightly longer.
- **No Advanced AI Features:** Does not yet provide personalized learning suggestions or adaptive learning paths.
- **Internet Dependency:** Requires stable internet for front-end updates, database access, and library downloads.
- **Manual Skill Updates:** Students must manually update their skill profiles to get accurate matches.
- **Limited Analytics:** Currently lacks detailed insights or reports on student performance and engagement.
- **No Gamification:** Does not have reward systems, badges, or incentives for active learning.
- **Limited Communication Features:** Chat is basic; lacks video calls or advanced collaboration tools.
- **No Offline Mode:** The system cannot be used without an internet connection.
- **Security Limitation:** Basic authentication only; advanced security like 2FA is not implemented.
- **Scalability Limitations:** May require database optimization for very large institutions.

CHAPTER-11

FUTURE ENHANCEMENT

The Student Skill Exchange Platform is a robust tool for peer-to-peer learning, but there are several opportunities to enhance its functionality and make it more intelligent, interactive, and scalable. Future enhancements can focus on improving **machine learning, accessibility, user engagement, analytics, and collaboration features**.

- **Advanced Machine Learning Algorithms:** Currently, the system uses TF-IDF and Cosine Similarity for peer recommendations. Future versions could incorporate **clustering algorithms, neural networks, or hybrid recommendation systems** to provide smarter, more adaptive peer matching. These enhancements could consider learning patterns, skill levels, and student interests to provide **more personalized recommendations**.

- **Mobile Application Development:** Introducing a **mobile version** (Android/iOS) will allow students to access the platform anytime, anywhere. Mobile notifications for peer requests, skill updates, and chat messages will improve **engagement and timely collaboration**.

- **Faculty Integration:** versions could include **faculty participation**, allowing mentors to guide discussions, approve requests, or suggest peer connections. This will create a **blended learning environment** combining student autonomy with expert guidance.

- **Gamification Features:** Adding **points, badges, leaderboards, or rewards** can motivate students to actively share skills and participate in learning sessions. Gamification encourages healthy competition and **increases user interaction** within the platform.

- **Advanced Analytics Dashboard:** Integrating a **dashboard with analytics** will allow students and administrators to track skill trends, popular learning topics, peer interactions, and engagement levels. This feature can **help identify skill gaps and plan future learning activities** efficiently.

- **Automated Learning Suggestions:** The platform could provide **recommendations for skill improvement courses, tutorials, or workshops** based on peer interactions and trending skills. This will help students **continuously improve their knowledge** without waiting for faculty intervention.

- **Enhanced Communication Tools:** Introducing features like **video chat, group discussion rooms, or file sharing** will expand collaborative capabilities and make online learning more interactive.

- **Offline Functionality:** Future updates could include **offline access**, allowing students to view resources or prepare skill lists even without internet connectivity.

- **Security Improvements:** Incorporating advanced security measures like **two-factor authentication, encrypted chat, and role-based access** will enhance the platform's safety and privacy.

- **Scalability and Cloud Deployment:** Hosting the system on a **cloud platform** can enable handling thousands of students simultaneously and ensure **high availability, performance, and data backup**.

- **Multi-Language Support:** Future versions of the system can support multiple languages so that students from different regions can use the platform comfortably. This will make the platform more accessible and inclusive for a larger group of users.

- **Skill Verification System:** The platform can include a feature where students can verify their skills through certificates, quizzes, or peer feedback. This will increase the reliability of skill information in the system.

- **Project Collaboration Module:** A dedicated module can be added where students can create project teams, assign roles, and collaborate on academic or technical projects directly through the platform.

- **Learning Resource Library:** Future updates could include a resource section where students can upload and share study materials, tutorials, notes, and project guides related to different skills.

- **Peer Rating and Feedback System:** Students can rate and give feedback after collaborating with peers. This feature will help maintain quality interactions and build trust within the platform.

- **Event and Workshop Notifications:** The platform can notify students about upcoming workshops, seminars, hackathons, or skill-development events related to their interests.

- **Integration with Learning Platforms:** The system can be integrated with online learning platforms such as MOOCs or e-learning portals to provide direct access to courses related to the student's skills.
- **AI-Based Skill Gap Detection:** An AI module can analyze student profiles and identify missing skills required for certain career paths or academic areas, helping students plan their learning goals.
- **Mentorship Program Integration:** Senior students or alumni can be connected with juniors through a mentorship feature, allowing guidance and knowledge sharing across different academic levels.
- **Automated Profile Suggestions:** The system can suggest improvements to student profiles by recommending additional skills, keywords, or descriptions that make their profiles more accurate and discoverable

CHAPTER-12

CONCLUSION

The **Student Skill Exchange Platform** successfully tackles a critical challenge in modern education: students' hesitation, fear of judgment, and reluctance to ask questions in traditional classroom settings. Many students possess strong technical or creative skills but often remain silent due to anxiety or lack of confidence. This platform bridges that gap, enabling a **safe and supportive environment** for knowledge exchange.

By leveraging **Machine Learning algorithms**, specifically TF-IDF and Cosine Similarity, the platform provides **automated, intelligent peer recommendations**. Students are connected with peers who share similar skills or possess complementary expertise, ensuring that collaboration is **effective, meaningful, and goal-oriented**. This automated matching system not only saves time but also enhances the **accuracy and relevance of peer connections** compared to manual methods.

The platform promotes **collaborative learning** through **requests, notifications, and chat sessions**, which allow students to communicate freely and confidently. By enabling peer-to-peer skill exchange, the system reduces dependency on faculty for doubt resolution, encouraging students to take initiative in their learning. This fosters **independence, proactive thinking, and problem-solving skills**, which are essential for both academic and professional growth.

Additionally, the platform demonstrates the **seamless integration of AI and web technologies**, showing how modern technological solutions can create **intelligent, interactive, and user-friendly learning environments**. Students not only gain practical experience in their respective domains but also develop **soft skills** such as communication, teamwork, and critical thinking. This helps in bridging the traditional gap between **theoretical knowledge and real-world applications**, preparing students for future challenges in higher education and the professional world.

The platform also encourages **inclusive learning**. Students who might otherwise feel isolated due to hesitation or communication barriers can now participate actively, share knowledge, and build confidence in a **peer-supported ecosystem**. It fosters an atmosphere of

collaboration, mutual growth, and skill development, making learning more engaging and accessible to all students.

- Addresses student hesitation, fear of judgment, and reluctance to ask doubts.
- Provides automated peer recommendations using TF-IDF and Cosine Similarity.
- Encourages collaborative learning through requests, notifications, and chat sessions.
- Reduces dependency on faculty, promoting independent and proactive learning.
- Bridges the gap between theoretical knowledge and practical application.
- Builds confidence and soft skills like communication, teamwork, and problem-solving.
- Creates an inclusive peer-supported ecosystem for all students.

In this project, machine learning techniques such as **TF-IDF and Cosine Similarity** are used to analyze student skill descriptions and identify similarities between profiles. These techniques help the system generate accurate peer recommendations based on the skills entered by users. The use of such algorithms improves the efficiency of the matching process and ensures meaningful connections between students.

The platform is designed with a simple and user-friendly interface so that students can easily register, update their skills, and explore recommended peers. Tools such as **Python, Scikit-learn, and Gradio** help in building the system and implementing the required functionalities effectively. These technologies support the development of a lightweight yet functional application

In essence, the Student Skill Exchange Platform is **more than a technical project**; it is a **transformative educational tool**. It empowers students to learn independently, collaborate effectively, and gain confidence in their abilities. By combining **advanced machine learning, efficient web technologies, and thoughtful system design**, the platform redefines peer learning and skill exchange, creating a **future-ready educational ecosystem** that is inclusive, intelligent, and empowering. It sets a strong foundation for **continuous innovation in education**, ensuring that every student has the opportunity to excel, collaborate, and succee

CHAPTER-13

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